

7 Collective Action Problems Quiz ANS

1. In the Prisoner's Dilemma as presented in the deck, what is the Nash equilibrium?

- A) Both players cooperate, which is also the dominant strategy for each of them
- B) Both players defect, although mutual cooperation would leave both better off
- C) One player cooperates while the other player chooses to defect in every round
- D) No equilibrium exists in the game, because it is strictly zero-sum between the players

ANS: B) Both players defect, although mutual cooperation would leave both better off

2. What is a Pareto improvement?

- A) A change that helps some or all parties without hurting anyone
- B) A state in which no player gains by changing strategy alone
- C) A change that helps one player at the expense of another player
- D) A change that equalizes every payoff regardless of efficiency

ANS: A) A change that helps some or all parties without hurting anyone

3. In the shopkeeper price-racing example, what is the Nash equilibrium?

- A) Both shopkeepers stop selling goods until demand returns
- B) Only one shopkeeper lowers prices while the other stays put
- C) Both shopkeepers keep their prices high to protect profit
- D) Both shopkeepers lower their prices, sacrificing profit

ANS: D) Both shopkeepers lower their prices, sacrificing profit

4. In the iterated Prisoner's Dilemma, why does defection remain the dominant strategy if agents know exactly how many times they will interact?

- A) Cooperation is always punished by the rules of the game itself in every round
- B) The payoffs change in the later rounds so that defection simply pays more
- C) Agents cannot remember what happened in any of the earlier rounds of the game
- D) Defection in the last round cannot be punished, and the logic unravels back through earlier rounds

ANS: D) Defection in the last round cannot be punished, and the logic unravels back through earlier rounds

5. What does the deck say can enable rational cooperation in repeated interactions?

- A) Reducing the number of players in the game to one
- B) Uncertainty about whether future interactions will occur
- C) Removing the ability to punish defection in any round
- D) Certainty that the final round is approaching soon

ANS: B) Uncertainty about whether future interactions will occur

6. In the corporate chatbot-release example, who is viewed as having "defected" in an iterated Prisoner's Dilemma?

- A) Companies that released chatbots without safety testing
- B) Regulators who proposed new rules in order to slow down the chatbots' release
- C) Companies that delayed their release in order to carry out additional safety testing
- D) Users who tried the chatbots and reported on their experience with them

ANS: A) Companies that released chatbots without safety testing

7. Which three developments does the deck compare when it calls military AI "the third revolution in warfare"?

- A) Radar, satellites, and uncrewed drones
- B) Gunpowder, nuclear weapons, and military AI
- C) Cavalry, artillery, and military aircraft
- D) Castles, cannons, and submarines

ANS: B) Gunpowder, nuclear weapons, and military AI

8. What does the security dilemma model describe?

- A) Mutual defensive concerns push nations to build up military capabilities, raising risks for all
- B) One nation's buildup always deters every other nation from arming itself at all
- C) Nations cooperate fully with one another because arms races are cheap to sustain
- D) A nation reduces its military whenever a rival reduces its own in response

ANS: A) Mutual defensive concerns push nations to build up military capabilities, raising risks for all

9. How does the deck define a flash war?

- A) A war fought exclusively in cyberspace by teams of human hackers working in shifts
- B) A short war fought only with conventional weapons by human soldiers on the ground
- C) A war that ends quickly because both sides promptly sign a treaty with each other
- D) A war triggered by autonomous AI systems that unfolds too quickly for human control

ANS: D) A war triggered by autonomous AI systems that unfolds too quickly for human control

10. In the autonomous economy model, what is the "defect" choice for companies?

- A) Maintaining human labor
- B) Sharing profits with workers
- C) Automating jobs using AI
- D) Lobbying for stricter regulation

ANS: C) Automating jobs using AI

11. Which of the following is NOT one of the five cooperation mechanisms listed in the deck?

- A) Iterated dominant-strategy elimination
- B) Institutions mediating interactions
- C) Group selection among competing groups
- D) Indirect reciprocity based on reputation

ANS: A) Iterated dominant-strategy elimination

12. Why does the deck say indirect reciprocity may not work with advanced AI?

- A) AIs interact with each other far too rarely to matter
- B) Indirect reciprocity requires that agents be genetic relatives
- C) AIs cannot form a reputation among one another at all
- D) Reciprocating with humans may offer the AI no upside

ANS: D) Reciprocating with humans may offer the AI no upside

13. Why does the deck argue that kin selection would not make AI kind toward humans?

- A) AIs and humans are close genetic relatives, so kin selection would already apply
- B) Kin selection applies only to formal institutions and treaties between nations
- C) Humans have very little information similarity to AIs, so the cost outweighs the benefit
- D) AIs are always perfectly selfless toward every other agent that they encounter

ANS: C) Humans have very little information similarity to AIs, so the cost outweighs the benefit

14. Which of the following is listed as a way institutions can fail or backfire?

- A) Eliminating all competitive pressures among organizations
- B) Making cooperation unnecessary for any group goal
- C) Perfect enforcement of rules at no cost to anyone
- D) Corruption, such as bribes and funding misappropriation

ANS: D) Corruption, such as bribes and funding misappropriation

15. In The Coup-Proofing Toolbox, what is "stovepiping"?

- A) Rotating officers so they cannot build trust
- B) Setting traps to purge would-be coup plotters
- C) Restricting information flow to make collusion harder
- D) Keeping your military in check with another military

ANS: C) Restricting information flow to make collusion harder

16. Which pair of assumptions does structural realism make about the international system?

- A) States answer to a world government and prioritize cooperation above everything else
- B) States operate in a self-help system with no higher arbiter and prioritize self-preservation
- C) States are irrational and always know the intentions of other states for certain
- D) States ignore power entirely, so domestic politics determines everything they do

ANS: B) States operate in a self-help system with no higher arbiter and prioritize self-preservation

17. What is a first strike advantage?

- A) An advantage that comes from delaying action indefinitely
- B) Whichever party initiates conflict gains the upper hand
- C) The party that strikes first is always punished severely
- D) A benefit that persists for decades after the initial strike

ANS: B) Whichever party initiates conflict gains the upper hand

18. What is issue indivisibility?

- A) The matter in dispute cannot be divided between the parties, so conflict is more likely
- B) The parties split a contested resource evenly between themselves without any conflict
- C) A dispute that cannot be explained by any known factor in the political science literature
- D) An issue over which every party already agrees completely from the start

ANS: A) The matter in dispute cannot be divided between the parties, so conflict is more likely

19. Which three conditions make up Lewontin's conditions for evolution by natural selection?

- A) Mutation, migration, and geographic isolation of groups
- B) Cooperation, competition, and extinction among populations
- C) Variation, retention, and differential fitness
- D) Selection, deception, and reciprocity within a species

ANS: C) Variation, retention, and differential fitness

20. Which two strategies does the deck suggest to avoid extreme AI selfishness?

- A) Ban all cooperation, and remove every form of oversight
- B) Remove all variation among AIs, and increase competition
- C) Reduce competitive pressures, and change what makes AIs fitter
- D) Increase competitive pressures, and reward the fastest AI

ANS: C) Reduce competitive pressures, and change what makes AIs fitter